Homework #5

14.1. A microstrip line is used as a feed line to a microstrip patch. The substrate of the line is alumina ($\varepsilon_r \simeq 10$) while the dimensions of the line are $w/h = 1.2$ and $t/h = 0$. Determine the effective dielectric constant and characteristic impedance of the line. Compare the computed characteristic impedance to that of a 50-ohm line.

解 $\varepsilon_r \approx 10$, $\frac{W}{h}=1.2$, $\frac{t}{h}=0$

故 $\varepsilon_{eff} = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}\left[1+12\frac{h}{W}\right]^{-1/2} = 6.257$

对于 $\frac{W_0}{h}>1$

$$Z_c = \frac{120\Omega}{\sqrt{\varepsilon_{reff}}\left[\frac{W_0}{h}+1.393+0.667\ln\left(\frac{W_0}{h}+1.444\right)\right]} = 44.4\Omega$$

$\delta Z = \left|\frac{Z_c - Z_s}{Z_s}\right| = 11.2\%$，即较50Ω低了11.2%

14.10. Design a rectangular microstrip patch with dimensions W and L, over a single substrate, whose center frequency is 10 GHz. The dielectric constant of the substrate is 10.2 and the height of the substrate is 0.127 cm (0.050 in.). Determine the physical dimensions W and L (in cm) of the patch, taking into account field fringing.

解: $f_r = 10GHz$, $\varepsilon_r = 10.2$, $h = 0.127cm$

$$W = \frac{c}{2f_r}\sqrt{\frac{2}{\varepsilon_r+1}} = 0.6339cm$$

$$\frac{W}{h} = \frac{0.6639cm}{0.127cm} = 4.991 > 1$$

故 $\varepsilon_{eff} = \frac{\varepsilon_r+1}{2} + \frac{\varepsilon_r-1}{2}\left(1+12\frac{h}{W}\right)^{-\frac{1}{2}} = 8.093$

$$L_{eff} = \frac{\lambda_g}{2} = \frac{\lambda_0}{2\sqrt{\varepsilon_{eff}}} = \frac{c}{2f_r\sqrt{\varepsilon_{eff}}} = 0.5273cm$$

$$\Delta L = 0.412h\frac{(\varepsilon_{reff}+0.3)\left(\frac{W}{h}+0.264\right)}{(\varepsilon_{reff}-0.258)\left(\frac{W}{h}+0.8\right)} = 0.0508cm$$

$$L = L_{eff} - 2\Delta L = 0.4254cm$$